

TITLE: DATA AGENT PROJECT

Author: Violet Lesunyane

Student Number: 2805001682

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1. INTRODUCTION

This project was developed as a test project to evaluate my ability to create a functional Data Agent using Databricks. The purpose was to assess my understanding of the project requirements, my ability to follow instructions, and my capability to successfully complete the assigned task.

2. OBJECTIVE

The objective of this project is to create a functional Retail Sales Data Agent that uses Artificial Intelligence to help business owners to better understand their companies through data-driven insights.

The Data Agent must be designed to:

- Analyse customer behaviour
- Identify why certain products perform better than others
- Detect underperforming products
- Improve sales performance analysis
- Support smarter business decision-making

The ultimate goal of the project is to help businesses use data and AI to make more informed strategic decisions, improve customer understanding, and drive business growth.

3. DATA DESCRIPTION

The provided data had the following data for me to Extract, Transform and Load on Genie Space:

Transaction ID

Transaction Date

Customer ID

Gender

Age

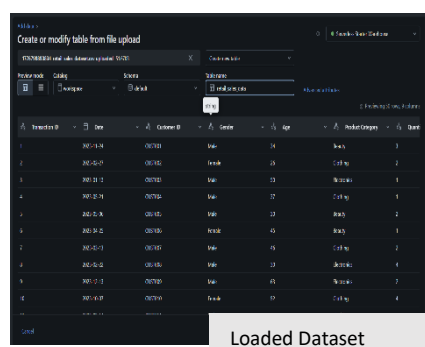
Product Category

Quantity

Price per Unit

Total Amount

4. EXTRACT > TRANSFORM > LOADING



Loaded Dataset

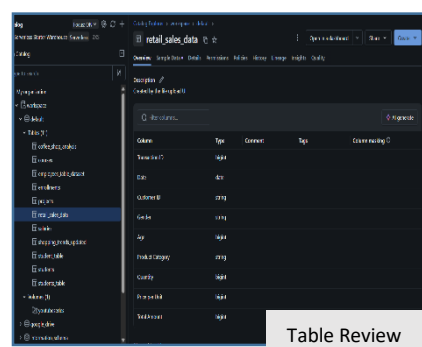


Table Review

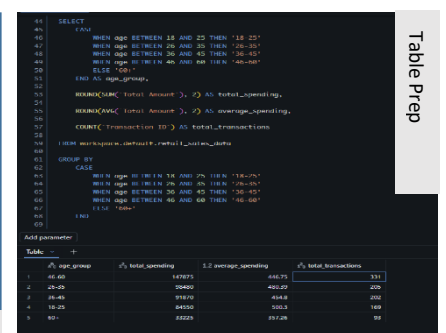
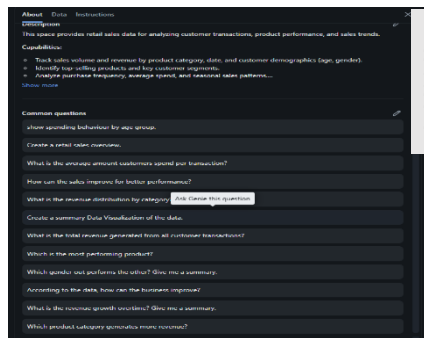
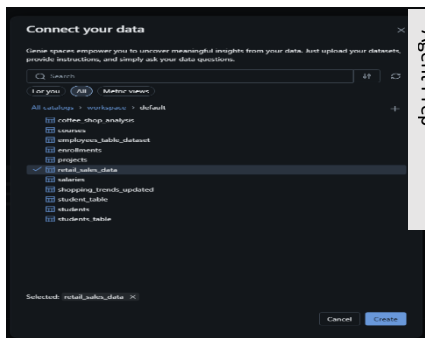
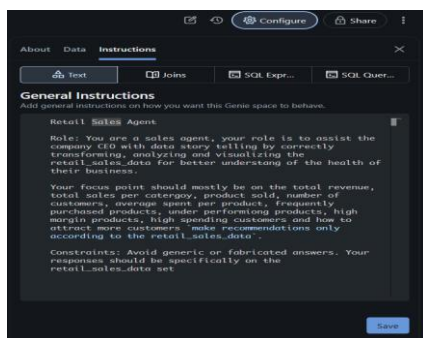


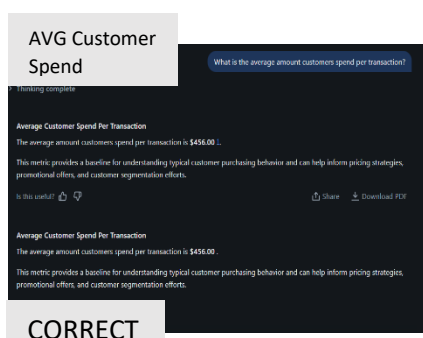
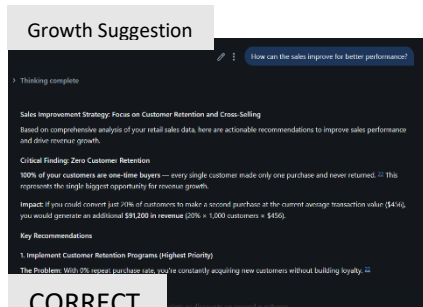
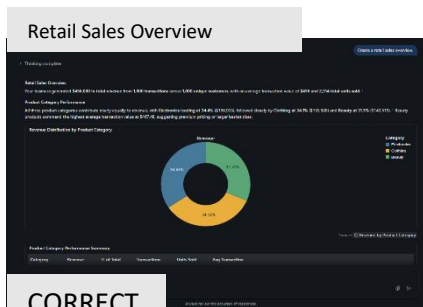
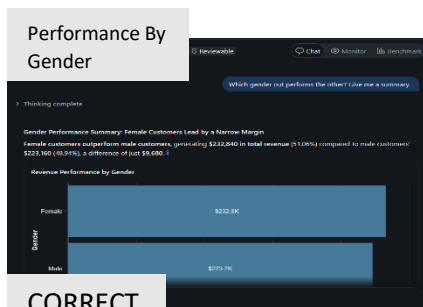
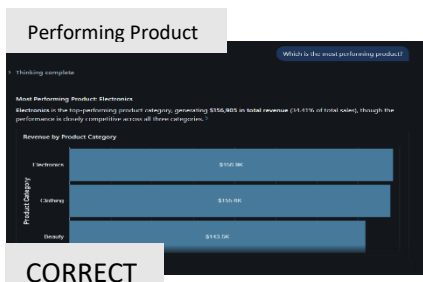
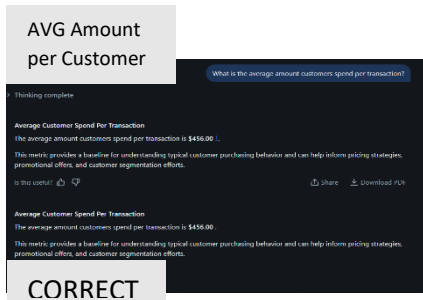
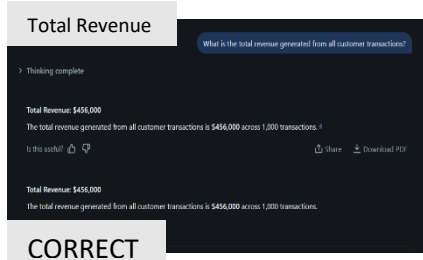
Table Prep

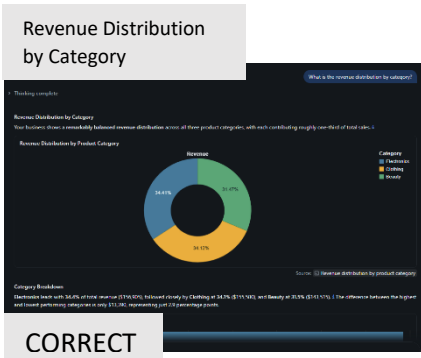


5. AGENT INSTRUCTIONS



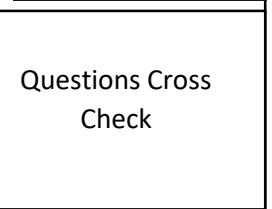
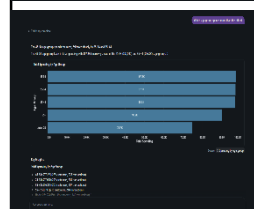
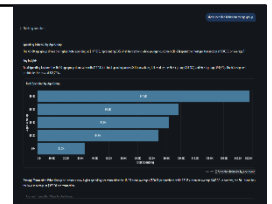
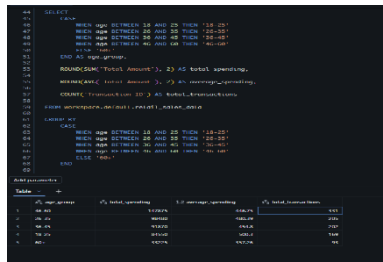
6. 10 QUESTIONS





Across all the questions I uploaded to the AI Agent, nine returned correct outputs, with the exception of the customer spending behaviour query. To resolve this issue, I reviewed the original query, copied it, and ran it in the Genie Space SQL environment, which produced more accurate results.

7. VALIDATION OF 3 ANSWERS



8. KEY INSIGHTS

The retail business is operationally functional but strategically underperforming with significant untapped growth potential.

8.1 The Core Problem

The business generated \$456,000 in revenue from 1,000 customers, but the critical weakness is zero customer retention every customer made exactly one purchase. This one-time buyer model severely limits growth and profitability, as acquiring new customers is typically 5-25 times more expensive than retaining existing ones.

8.2 What's Working

Balanced Product Portfolio: All three categories (Electronics, Clothing, Beauty) contribute roughly equally to revenue (31-34% each), demonstrating diversified income streams and reducing dependency risk.

Strong Average Transaction Value: At \$456 per transaction, customers are willing to spend meaningfully when they do purchase, with Beauty products commanding the highest value at \$467.481.

Valuable Core Demographic: The 46-60 age group represents your most reliable customer base, contributing 32% of total revenue (\$147,875).

8.3 What's Not Working

No Customer Loyalty: The complete absence of repeat purchases indicates fundamental issues with customer experience, engagement, or retention strategy.

Revenue Volatility: Monthly sales swing wildly from \$23,620 to \$53,150, suggesting inconsistent marketing, inventory management, or seasonal planning.

Underutilized High-Value Segments: Young adults (18-25) spend the most per transaction (\$500.30) but represent only 17% of customers, while seniors (60+) are barely engaged with just 93 customers.

9. BUSINESS RECOMMENDATIONS

Immediate Priority: Implement a customer retention program (loyalty rewards, email marketing, personalized follow-ups) to convert one-time buyers into repeat customers. Even a modest 20% repeat purchase rate could add at least \$90,000+ in annual revenue without new customer acquisition costs.

Growth Opportunity: Target the 18-25 demographic with digital-first marketing and expand senior outreach to capture underserved high-potential segments.

Stabilization Need: Address monthly revenue volatility through consistent promotional calendars, better inventory planning, and understanding what drove the companies best and worst months.

In Conclusion

The business is viable, with solid fundamentals but is operating at a fraction of potential capacity due to the absence of customer retention. The business is currently in "acquisition-only mode" when it should be in "acquire and retain mode." Shifting focus from purely transactional relationships to building customer loyalty will be the difference between stagnant performance and sustainable growth. The data shows customers are willing to spend you just need to give them reasons to come back.

10. CONCLUSION

10.1 What Worked

One of the key things that worked well during this project was understanding the importance of querying and analysing the dataset before connecting it to Genie Space. Preparing and understanding the data beforehand helped produce more accurate responses and improved the overall quality of the AI agent's outputs. AI agents and Large Language Models such as OpenAI ChatGPT perform best when they are provided with sufficient and accurate context. Without proper context and structured data, the agent becomes less effective and may produce unreliable results.

10.2 What Was Hard

I did not experience major difficulties during the project; however, I would still like to continue improving and mastering my coding languages and technical skills to achieve even better analytical and AI-driven results in future projects.

10.3 What I Would Change

One thing I would change is doubting my own abilities. This project helped me realise that data science is a valuable and rewarding journey. The ability to use data to assist businesses in making informed, data-driven decisions is something I find highly meaningful and motivating. As I continue growing as a Data Scientist, I aim to keep learning, improving my technical skills, and developing solutions that create real business value.